

Study Questions: Thinking Like an Inventor

Analytical Questions

1. Active Inference describes cognition as "inference over hidden causes." Explain what this means in the context of an inventor diagnosing why a new type of bread dough keeps collapsing during baking. What are the hidden causes? What are the observable symptoms? How does the inventor's generative model determine which causes they consider?
2. Bela Barenzyi reframed automotive safety from "rigidity protects" to "controlled deformation protects." Analyze this reframing using Active Inference terminology. What changed in the generative model's structure? Why was this a structural change rather than a parameter update? What made the old model resistant to change?
3. Compare mental simulation and physical prototyping as methods for evaluating candidate solutions. What are the advantages of each? Under what conditions does mental simulation give misleading results? Use the concept of generative model accuracy to frame your answer.
4. Functional fixedness and Einstellung (the set effect) are two cognitive biases that impede inventive thinking. Define each, give an invention-relevant example of each, and explain how Active Inference's emphasis on model comparison could help overcome them.
5. The development of penicillin required three different types of cognition: Fleming's perceptual recognition, Chain's mechanistic reasoning, and Florey's applied/clinical reasoning. Using Active Inference, explain why a single inventor's generative model was unlikely to encompass all three cognitive modes. What does this imply about the cognitive structure of inventive teams?
6. Counterfactual reasoning allows inventors to imagine scenarios that do not yet exist. How does Active Inference support counterfactual reasoning through the generative

model? What are the limits of counterfactual reasoning when the inventor's model is incomplete or inaccurate?

7. The "Five Whys" technique traces symptoms backward to root causes. Explain how this technique relates to the depth of the inventor's generative model. What happens when the inventor reaches a "why" they cannot answer? What strategies can they use to extend their causal chain beyond their current knowledge?
8. James Dyson tested over 5,000 prototypes of his cyclonic vacuum cleaner. In Active Inference terms, what was he doing cognitively during this process? Was each prototype testing the same generative model with different parameters, or was he changing model structure along the way? What role did each failed prototype play in updating his cognitive model?
9. Explain the concept of Bayesian model comparison as it applies to an inventor evaluating three competing theories about why their solar panel design underperforms. How should the inventor weight evidence for and against each theory? What observation would be most decisive in distinguishing between the theories?
10. Problem abstraction (moving from "how do I build a better mousetrap" to "how do I keep mice out of the pantry") opens new solution spaces. Analyze why the abstract framing is more generative. At what point does abstraction become too broad to be useful? Where is the optimal level of abstraction for inventive cognition?

Applied Questions

1. Choose a problem you encounter daily. Trace the causal chain from symptom to root cause using the Five Whys technique. Then identify at least one alternative causal chain that could produce the same symptom. Which causal chain suggests the more actionable intervention?
2. Take your current invention idea and reframe it using the inversion technique: instead of asking "how do I solve this problem," ask "how would I create this problem on purpose?" What new causal insights does the inversion reveal?

3. Identify one constraint you have been assuming about your invention that might be self-imposed rather than real. If you removed this constraint, what new solutions would become possible? Is the constraint truly necessary?
4. Run a mental simulation of your invention in use. Walk through the user experience step by step, from first encounter to regular use. Where does your mental simulation become uncertain or fuzzy? These are the areas of your generative model that most need testing through prototyping.
5. Describe a situation where you experienced functional fixedness — where you could not see a familiar object or process being used in a new way. What broke the fixation? How could you have broken it sooner?
6. Generate three radically different solutions to your invention problem. The first should be within your domain expertise. The second should borrow from a completely different domain. The third should violate one of your assumed constraints. Compare them: which has the most creative potential?
7. Choose an invention from another field (different from yours) and reconstruct the inventor's likely cognitive process. What causal model did they build? What reframing was required? What counterfactual reasoning did they use? Base your analysis on what you know about the invention's history.
8. Assess your own cognitive flexibility using the indicators from the lab (generating multiple explanations, considering alternatives, changing your mind, questioning assumptions, imagining different perspectives). Which indicators scored lowest? Design a specific practice to improve your weakest area.
9. Consider a design decision in your invention where you are choosing between two approaches. For each approach, predict three consequences — one immediate, one medium-term, and one long-term. Which approach has better consequences at each timescale? Does the best short-term choice match the best long-term choice?
10. Write a counterfactual history of a well-known invention. What if the key reframing had not occurred? For example, what if automotive safety engineers had never conceived of

crumple zones? How would cars be different today? What does this counterfactual tell you about the importance of cognitive reframing in the trajectory of technology?